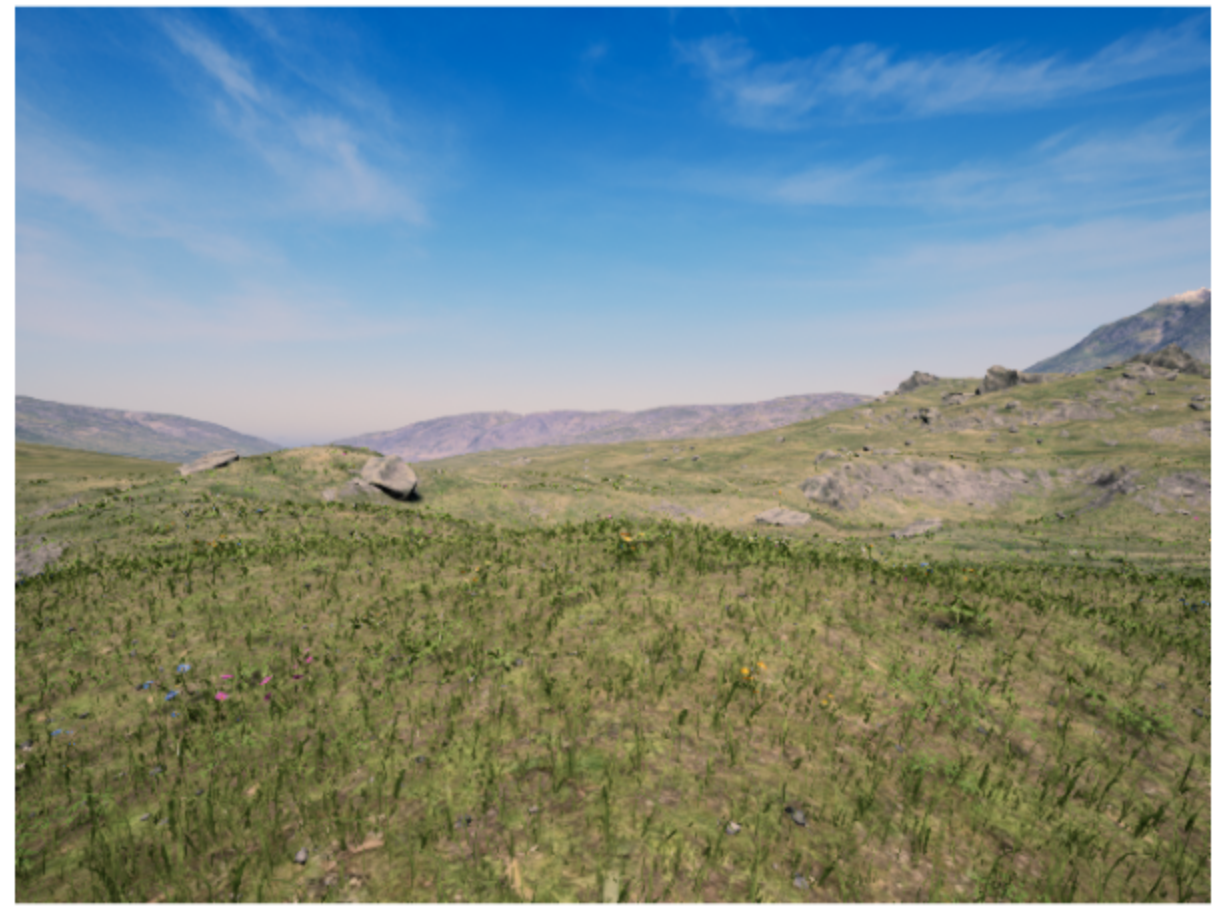
RflySim for Real4Real

Two stock PX4 routes completed with confirmed landing and disarming.

Native RGB capture worked at 640 × 480 and about 10 Hz.

Reuse control integration and a stock visual baseline. Defer calibrated reconstruction rendering.



Inputs and component boundaries

Input: Free v5.01 ISO, stock Grasslands, Iris model 0, frozen NED route. No policy weights or real reconstruction.

Python sends setpoints to PX4. CopterSim advances dynamics. PX4 returns estimates and mode/arm ACKs.

UE4 renders the camera. A separate receiver saves images and raw camera/body metadata. Calibrated association is still missing.

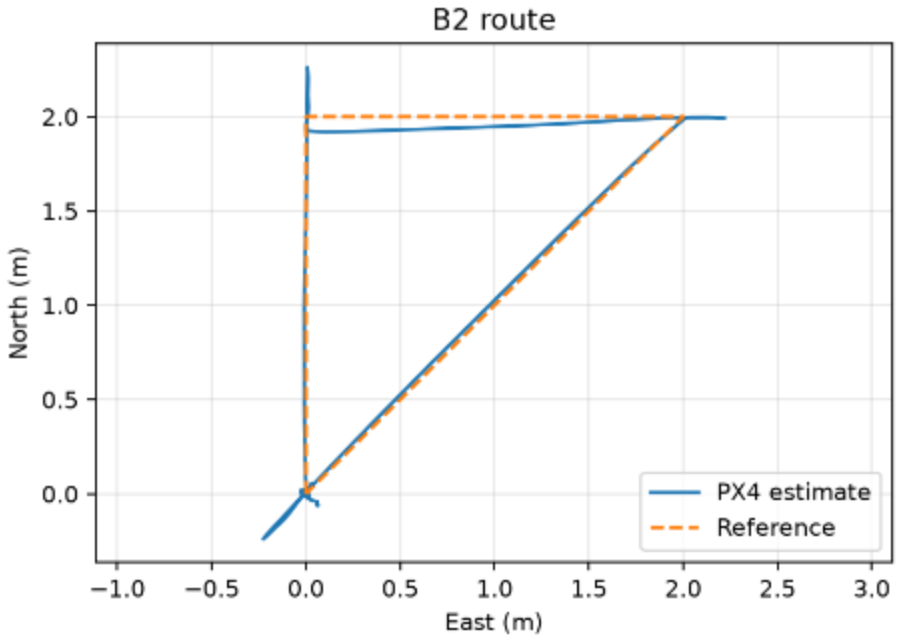
Installation and graphics now work

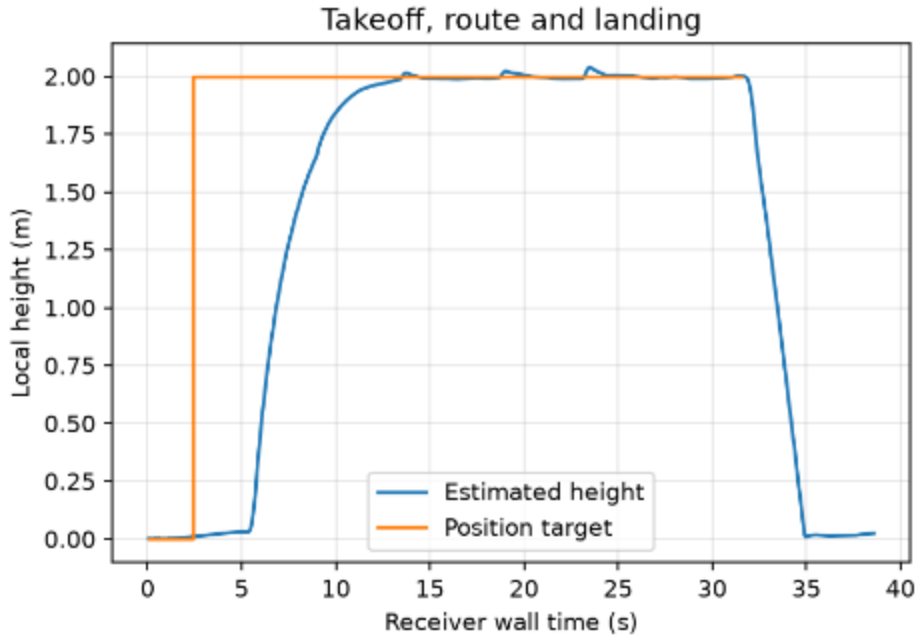
The bundled Epic prerequisite installer completed with exit 0. Correct renderer working directory resolved startup in the retest.

RflySim3D selected NVIDIA L4 through D3D11. UE4 4.27.2 loaded Grasslands.

Server 2025 and WSL1 worked on this host. Vendor documentation names Windows 10/11. Stock execution needed no licensed MATLAB.

Both flights completed the route



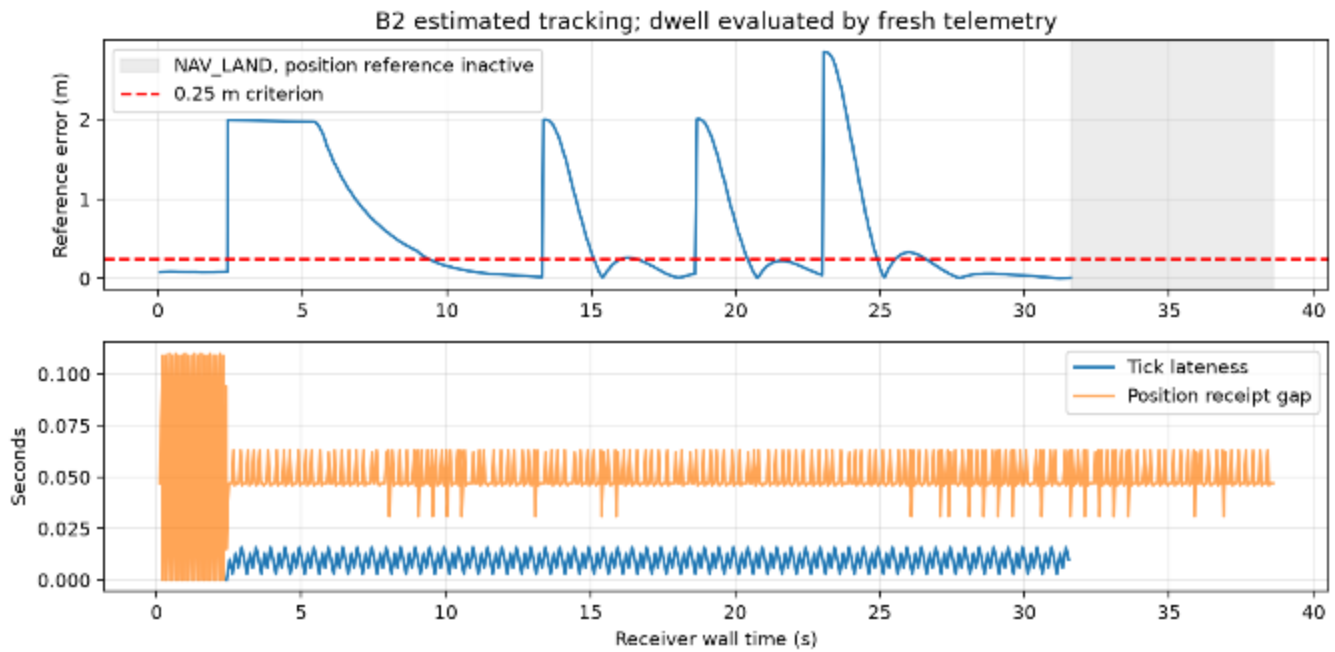


Both full-route passes used fresh PX4 estimates. B1: 38.750 s. B2 with capture: 39.219 s.

Route in NED metres: (0,0,-2), (2,0,-2), (2,2,-2), (0,0,-2), land. Each waypoint required ≤0.25 m for 2 s.

The pass uses fresh PX4 estimates. Independently calibrated truth outcome remains blocked.

Capture comparison stays limited



Mean setpoint interval: 100.022 ms without capture, 100.034 ms with capture.

Maximum tick lateness: 12 ms versus 16 ms. Neither run had a tick more than 100 ms late. State gaps peaked at 125 ms and 110 ms.
Cleanup workers hit 15 s despite confirmation summaries. Raw cleanup logs are empty. Queue repair needs retesting.

Two sequential runs do not isolate capture overhead. Cleanup logging is incomplete.

Flight and observations share a run

B2 saved 365 mounted-camera RGB frames while the vehicle completed the route.

Trace records include setpoints, ACKs, mode/armed state, estimates, body truth, camera metadata and frame IDs.

Acquisition-time camera pose remains unknown. RGB recording did not drive the waypoint controller.



Camera geometry exposed our defect

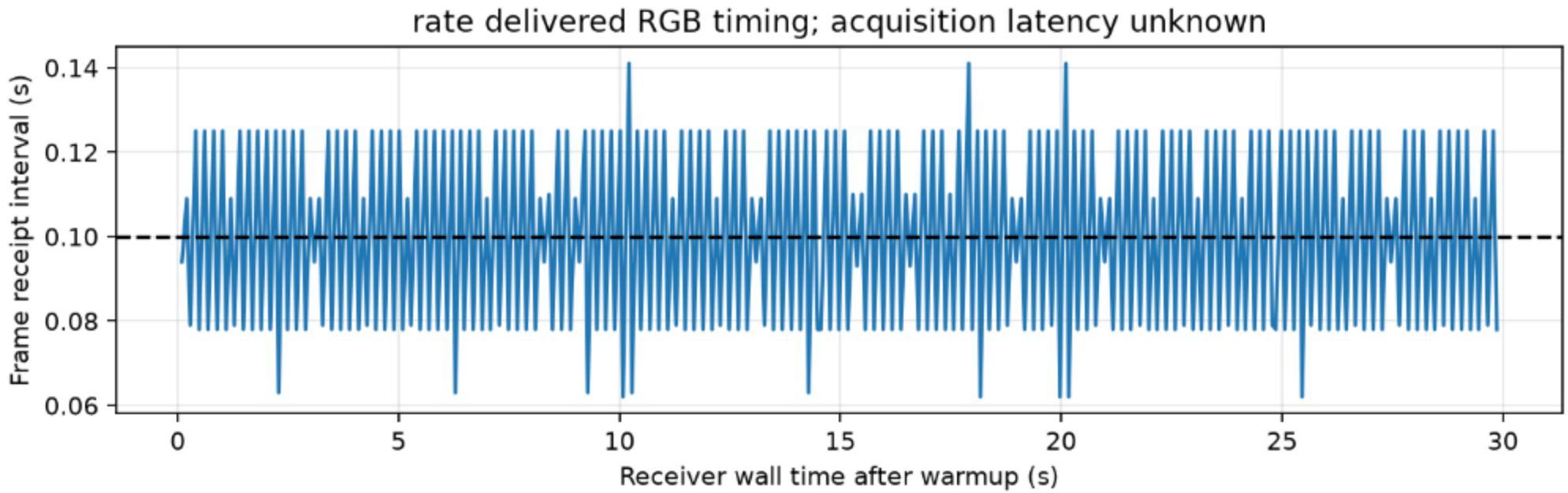
A1 and A2 saved 260 and 261 frames. A1 captured 25.797 seconds, short of the requested 26.

Our adapter converted degrees to radians for a degree field. The intended 15° yaw and 10° pitch were not executed.

Local duration and degree fixes passed helper tests. They need a simulator retest. Pose repeatability and projection checks remain blocked.



Native capture meets the rate request

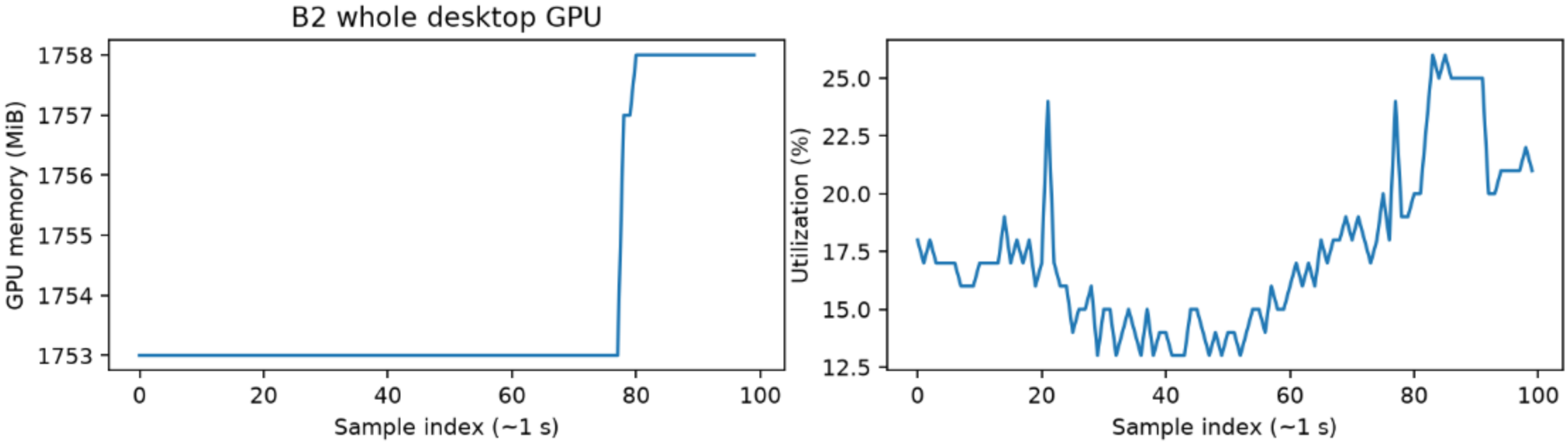


The 35 s test included 5 s warmup and 300 post-warmup frames. Delivered record rate was 10.013 Hz.

No duplicate native timestamps or post-warmup receipt gaps above 150 ms occurred in the saved rate sequence.

A2 first image had an old-looking clock value before a large jump. Clock ownership and acquisition latency remain unresolved.

Resource measurements fit this workload



Peak GPU allocation in the sample windows: 1,761 MiB and 1,758 MiB. Mean utilization: 16.93% and 17.47%.

The host had 8 logical CPUs, 32 GB RAM and a 400 GB SSD. Driver 596.86, reported L4 memory 23,034 MiB.

Polling includes preflight, Moonlight and desktop. B2 sampling ended mid-flight. Minimum VRAM and renderer-only throughput remain unmeasured.

Reconstruction and Gaussian boundaries

Real photos plus calibrated poses: no verified direct reconstruction loader. Cooked UE mesh/XML import is documented, with Free eligibility unresolved.

External trajectory replay: a conditional small adapter after camera calibration, with an existing Gaussian scene and renderer.

Native RGB replacement or a visual closed loop: unresolved supported contract and substantial integration. Collision/depth and clock ownership need separate services.

Scoped R4R decision

Reuse as a dynamics/control host and keep a native RGB baseline for integration experiments.

Defer calibrated replay and R4R reconstruction rendering until corrected rotations, projection/extrinsics and acquisition timing are verified.

Indoor navigation, residential/campus views below 100 m, sparse capture and racing remain research questions. C is blocked by missing compatible released weights and task adapter.

Preservation and the next review

Three ZIP checkpoints passed hashes and internal manifests on the Mac. Main telemetry, 1,237 frames, configs, executed sources and B1 native flight log are preserved.

AirGPU is verified Off. Final transcript, B2 native flight log and the strict A1 retest remain on its stopped disk.

Read Link/FlightEvaluation, camera handshake, SDK degree versus yaw-radian conventions, and the clock diagnostics. Next work is corrected calibrated A and missing-file recovery.